

## 14. LOG ANALYSIS FOR DETECTION AND RESPONSE

DATE: 22/04/2026

### Aim:

To understand the fundamentals of log analysis and its role in effective security detection and response.

### Procedure:

1. Log into TryHackMe and open the Intro to Log Analysis room.
2. Study the fundamentals of log analysis and its role in detection and response.
3. Understand different log types such as system, network, and application logs.
4. Learn best practices for collecting, storing, and analyzing logs.
5. Explore essential tools used for monitoring and identifying suspicious activity.
6. Complete the tasks and answer the questions in the room.
7. Review the summary and confirm completion.

### Tasks:

← Back to all walkthroughs

### Intro to Log Analysis

An intro to log analysis, best practices, and essential tools for effective detection and response.

📶 🌐 ⌚ 60 min 👤 42,133 🏆

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Room completed (100%)

- Task 1 ✓ Introduction
- Task 2 ✓ Log Analysis Basics
- Task 3 ✓ Investigation Theory
- Task 4 ✓ Detection Engineering
- Task 5 ✓ Automated vs. Manual Analysis
- Task 6 ✓ Log Analysis Tools: Command Line
- Task 7 ✓ Log Analysis Tools: Regular Expressions
- Task 8 ✓ Log Analysis Tools: CyberChef
- Task 9 ✓ Log Analysis Tools: Yara and Sigma
- Task 10 ✓ Conclusion

### TASK 1 - Introduction

Answer the questions below

I'm ready to proceed!

No answer needed

✓ Correct Answer

## TASK 2 - Log Analysis Basics

Answer the questions below

I understand the basics of logs and I'm ready to proceed!

No answer needed

✓ Correct Answer

## TASK 3 - Investigation Theory

Answer the questions below

What's the term for a consolidated chronological view of logged events from diverse sources, often used in log analysis and digital forensics?

Super Timeline

✓ Correct Answer

Which threat intelligence indicator would 5b31f93c09ad1d065c0491b764d04933 and 763f8bdbc98d105a8e82f36157e98bbe be classified as?

File Hashes

✓ Correct Answer

## TASK 4 - Detection Engineering

Answer the questions below

What is the default file path to view logs regarding HTTP requests on an Nginx server?

/var/log/nginx/access.log

✓ Correct Answer

A log entry containing %2E%2E%2F%2E%2Fproc%2Fself%2Fenvron was identified. What kind of attack might this infer?

Path Traversal

✓ Correct Answer

## TASK 5 – Automated vs Manual Analysis

Answer the questions below

A log file is processed by a tool which returns an output. What form of analysis is this?

Automated

✓ Correct Answer

An analyst opens a log file and searches for events. What form of analysis is this?

Manual

✓ Correct Answer

## TASK 6 – Log Analysis Tools:Command Line

```
(JackSparrow@Captain)-[~/Desktop]
$ cut -d' ' -f7 apache-1691435735822.log | sort |
uniq
/about.php
/admin.php
/contact.php
/index.php
/index.php?flag=c701d43cc5a3acb9b5b04db7f1be94f6
/login.php
```

```
(JackSparrow@Captain)~[~/Desktop]
$ grep ' 200 ' apache-1691435735822.log | wc -l
52
```

```
(JackSparrow@Captain)~[~/Desktop]
$ awk '{print $1}' apache-1691435735822.log | so
rt | uniq -c | sort -nr | head -1
8 145.76.33.201
```

```
(JackSparrow@Captain)~[~/Desktop]
$ grep '110.122.65.76' apache-1691435735822.log
| grep '/login.php'
110.122.65.76 - - [31/Jul/2023:12:34:40 +0000] "GE
T /login.php HTTP/1.1" 200 9876 "Mozilla/5.0 (Wind
ows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML
, like Gecko) Chrome/95.0.4638.110 Safari/537.36"
```

#### Answer the questions below

Use `cut` on the `apache.log` file to return only the URLs. What is the flag that is returned in one of the unique entries?

✓ Correct Answer



In the `apache.log` file, how many total HTTP 200 responses were logged?

✓ Correct Answer



In the `apache.log` file, which IP address generated the most traffic?

✓ Correct Answer



What is the complete timestamp of the entry where `110.122.65.76` accessed `/login.php`?

✓ Correct Answer



## TASK 7 – Log Analysis Tools : Regular Expressions

#### Answer the questions below

How would you modify the original `grep` pattern above to match blog posts with an ID between 20-29?

✓ Correct Answer



What is the name of the filter plugin used in Logstash to parse unstructured log data?

✓ Correct Answer

## TASK 8 – Log Analysis Tools : CyberChef

**Recipe**

Regular expression

Built in regexes

IPv4 address

Reqex

```
(?:\d{1,3}\.\d{1,3}\.\d{1,3}\.\d{1,3})
```

☒ Case insensitive

☒ ^ and \$ match at newlines

☐ Dot matches all ☐ Unicode support

☐ Astral support ☐ Display total

Output format

List matches

STEP **BAKE!** Auto Bake

**Input**

```
143.110.222.166 - - [27/Sep/2023:00:14:08 +0000] "GET / HTTP/1.1" 401 677 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 16_1 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/16.1 Mobile/15E148 Safari/604.1"
45.79.172.21 - - [27/Sep/2023:00:40:38 +0000] "GET / HTTP/1.1" 401 5196 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 13_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/108.0.0.0 Safari/537.36"
34.76.158.233 - - [27/Sep/2023:00:55:40 +0000] "GET / HTTP/1.1" 401 733 "-" "python-requests/2.31.0"
74.59.70.120 - - [27/Sep/2023:01:00:11 +0000] "POST
```

**File details**

Name: access.log  
Size: 19,051 bytes  
Type: unknown

**Output**

```
43.133.134.155
115.0.0.0
43.133.134.155
212.14.17.145
43.133.134.155
43.133.134.155
43.133.134.155
43.133.134.155
43.133.134.155
43.133.134.155
43.133.134.155
```

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**Recipe**

Regular expression

Built in regexes

User defined

Reqex

```
(POST|GET) /.*=
```

☒ Case insensitive

☒ ^ and \$ match at newlines

☐ Dot matches all ☐ Unicode support

☐ Astral support ☐ Display total

Output format

List matches

STEP **BAKE!** Auto Bake

**Input**

```
143.110.222.166 - - [27/Sep/2023:00:14:08 +0000] "GET / HTTP/1.1" 401 677 "-" "Mozilla/5.0 (iPhone; CPU iPhone OS 16_1 like Mac OS X) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/16.1 Mobile/15E148 Safari/604.1"
45.79.172.21 - - [27/Sep/2023:00:40:38 +0000] "GET / HTTP/1.1" 401 5196 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 13_1) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/108.0.0.0 Safari/537.36"
34.76.158.233 - - [27/Sep/2023:00:55:40 +0000] "GET / HTTP/1.1" 401 733 "-" "python-requests/2.31.0"
74.59.70.120 - - [27/Sep/2023:01:00:11 +0000] "POST
```

**File details**

Name: access.log  
Size: 19,051 bytes  
Type: unknown

**Output**

```
GET /dns-query?name=dnsscan.shadowserver.org&type=
GET /?XDEBUG_SESSION_START=
GET /VEhNe0NZQkVSQ0hFRl9XSvpBUkR9==
GET /viwwwsogou?op=8&query=
```

**Recipe**

**From Base64**

Alphabet  
A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

**Input**

VEhNe0NZQkVSQ0hFR19XSvpBUkR9

**Output**

THM{CYBERCHEF\_WIZARD}

**Recipe**

**From Base64**

Alphabet  
A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

**Regular expression**

Built in regexes  
MAC address

Regex:  
[A-Fa-f\d]{2}(?:[:-][A-Fa-f\d]{2}){5}

☒ Case insensitive

**Input**

MDgtMkUtOUeItNEItN0ZFLTYxMEEtM0MtOEItNUQtNkFGLTkYMEItMU  
UtNkMtN0QtNEZFLTgzMEMtNEItN0EtOUQtMkNfLTe1MEQtNkYtOEMt  
MUUtM0JELTc0MEUtN0QtMkItNtUMtOUFFLTyWMEYtNUMtM0QtOEItNE  
FFLTi3MUeTOUQtMkMtM0UtN0JGLTU4MUItOEUtNEYtN0EtNkRFLTiW  
MUMtN0EtM0UtNUYtOUNGLTQ2MUQtNkItNEEtOEYtMkJFLTA5MUUtNE  
YtOUQtNUMtM0FFLTy3MUYtM0EtNkItMkQtOEZFLTyMkEtMUQtOEYt  
OUMtNUJFLTc2MkItMEMtNUEtM0QtNEZFLTgxMkMtOUItN0UtOEQtMU  
FGLTM0MkQtOEMtNkYtNUUtMERFLTE0MkUtN0YtNEItNkEtOUNFLTg5  
MkYtNkUtMEQtMUMtM0ZFLTU3M0EtMEQtOEYtN0UtNkJGLTQyM0ItOU  
MtMkUtMUytNEFFLTyzM0MtOEItNUQtMEYtN0RFLTi5M0QtN0EtMUMt  
MkItOUZFLTUxM0UtNkYtNEEtOUMtNUNFLTM4M0YtNUUtN0ItOEQtMk  
EGiTV0MEtN0QtNUIIIMIMtNLA5iTTtMETNlUMtOEYtM0EtOUBGiTOA

**File details**

Name: encodedflag.txt  
Size: 4,344 bytes  
Type: unknown

**Output**

08-2E-9A-4B-7F-61

**Answer the questions below**

Locate the "loganalysis.zip" file under `/root/Rooms/introloganalysis/task8` and extract the contents.

No answer needed ✓ Correct Answer

Upload the log file named "access.log" to CyberChef. Use regex to list all of the IP addresses. What is the full IP address beginning in 212?

212.14.17.145 ✓ Correct Answer

Using the same log file from Question #2, a request was made that is encoded in base64. What is the decoded value?

THM{CYBERCHEF\_WIZARD} ✓ Correct Answer

Using CyberChef, decode the file named "encodedflag.txt" and use regex to extract by MAC address. What is the extracted value?

08-2E-9A-4B-7F-61 ✓ Correct Answer

## TASK 9 – Log Analysis Tools : Yara and Sigma

Answer the questions below

What languages does Sigma use?

YAML

✓ Correct Answer

What keyword is used to denote the "title" of a Sigma rule?

title

✓ Correct Answer

What keyword is used to denote the "name" of a rule in YARA?

rule

✓ Correct Answer

## TASK 10 - Conclusion

Answer the questions below

Click and continue learning!

No answer needed

✓ Correct Answer

## Result:

Thus the TryHackMe room "Intro to Log Analysis" was completed successfully.